

LINKÖPING MALMEN, Sweden

WMO: 025620

Lat: 58.402N

Lon: 15.526E

Elev: 94

StdP: 100.20

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.6	-12.9	-18.4	0.7	-16.1	-14.9	1.0	-11.9	12.2	5.1	10.9	5.2	1.1	250	0.617

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.7	26.9	18.4	25.0	17.8	23.2	16.9	19.8	24.3	18.6	23.1	17.7	21.8	3.4	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.2	13.2	21.4	17.0	12.3	20.4	16.0	11.5	19.5	56.9	24.6	53.1	23.2	50.1	21.8	23.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.1	8.9	7.8	DB	-19.7	29.2	4.3	2.3	-22.8	30.9	-25.3	32.2	-27.7	33.5	-30.9	35.2	(4)
(5)			WB	-19.9	21.3	4.3	1.3	-23.0	22.2	-25.5	23.0	-27.9	23.7	-31.0	24.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.8	-2.0	-2.0	0.8	5.7	10.4	14.3	16.9	16.1	12.0	6.9	2.7	-0.6	(6)	
	DBStd	7.78	4.94	5.07	4.26	3.55	3.54	2.80	2.86	2.72	3.05	3.57	3.90	5.08	(7)	
	HDD10.0	1841	373	336	286	136	38	2	0	0	16	106	221	329	(8)	
	HDD18.3	4244	632	569	544	379	247	126	65	80	192	354	470	587	(9)	
	CDD10.0	673	0	0	0	7	49	129	213	189	74	11	1	0	(10)	
	CDD18.3	34	0	0	0	0	1	4	19	11	0	0	0	0	(11)	
	CDH23.3	368	0	0	0	1	14	58	207	87	2	0	0	0	(12)	
	CDH26.7	59	0	0	0	0	1	6	41	11	0	0	0	0	(13)	
(14) Wind	WSAvg	3.6	3.8	4.0	4.0	3.6	3.5	3.5	3.2	3.2	3.6	3.7	3.7	4.0	(14)	
(15) Precipitation	PrecAvg	552	39	26	27	34	39	39	70	75	55	46	58	44	(15)	
	PrecMax	710	71	73	59	71	86	71	111	151	108	120	119	120	(16)	
	PrecMin	393	7	10	1	1	9	11	24	16	18	4	24	7	(17)	
	PrecStd	97	18	17	19	17	25	16	23	42	25	31	28	28	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.3	9.1	13.6	20.2	25.2	27.5	30.1	27.9	23.2	16.7	12.2	10.5	(19)	
		MCWB	7.2	7.3	8.2	12.7	16.8	18.1	19.4	18.4	17.2	13.8	10.8	9.3	(20)	
	2%	DB	6.4	7.0	10.8	17.2	22.3	25.0	27.6	25.7	20.7	14.7	10.6	8.3	(21)	
		MCWB	5.3	5.5	6.9	10.8	14.5	17.3	18.9	18.2	15.9	12.7	9.4	7.2	(22)	
	5%	DB	5.1	5.8	8.5	14.6	20.0	22.8	25.7	23.8	18.7	13.4	9.4	7.0	(23)	
		MCWB	4.0	4.5	5.7	9.5	13.7	16.0	18.1	17.5	15.0	11.8	8.4	6.0	(24)	
	10%	DB	3.9	4.5	6.8	12.4	17.6	20.6	23.7	21.9	17.0	12.1	8.1	5.6	(25)	
		MCWB	3.0	3.3	4.6	8.1	12.1	14.8	17.4	16.7	14.0	10.7	7.2	4.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.3	7.3	9.1	13.8	18.0	19.6	21.4	20.8	18.3	14.5	11.2	9.6	(27)	
		MCDB	8.1	8.5	12.6	19.0	23.8	25.0	27.1	25.0	21.7	16.1	12.0	10.5	(28)	
	2%	WB	5.5	5.8	7.2	11.4	15.9	17.9	20.2	19.4	16.6	13.1	9.5	7.4	(29)	
		MCDB	6.2	6.8	9.8	15.5	20.3	22.9	25.4	23.0	19.7	14.3	10.3	8.1	(30)	
	5%	WB	4.2	4.6	6.0	9.9	14.2	16.7	19.0	18.3	15.4	12.0	8.5	6.1	(31)	
		MCDB	4.9	5.6	8.1	13.9	18.6	21.6	24.0	22.0	18.0	13.1	9.3	6.9	(32)	
	10%	WB	3.1	3.4	4.8	8.5	12.8	15.5	17.9	17.4	14.3	10.9	7.2	4.8	(33)	
		MCDB	3.8	4.3	6.6	11.8	17.0	19.8	22.5	21.0	16.6	12.0	7.9	5.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	4.6	5.4	7.2	9.4	10.2	9.7	9.7	9.2	8.1	6.2	4.4	4.6	(35)	
		MCDBR	5.0	5.3	9.8	13.8	14.5	13.9	13.7	12.2	10.8	7.0	5.5	5.2	(36)	
	5% WB	MCWBR	4.6	4.4	6.6	8.2	8.0	7.1	6.2	5.9	6.4	5.4	4.9	4.9	(37)	
		MCWBR	4.9	5.3	9.1	12.1	12.2	11.9	11.9	10.2	9.7	6.6	5.3	5.2	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.279	0.317	0.336	0.374	0.370	0.359	0.385	0.376	0.345	0.335	0.306	0.266	(40)	
		taud	2.239	2.299	2.358	2.318	2.392	2.445	2.393	2.414	2.487	2.429	2.299	2.176	(41)	
	Ebn at Noon		558	710	809	832	863	878	843	823	798	686	519	439	(42)	
		Edh at Noon	46	70	88	108	107	103	106	97	78	62	44	35	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.35	0.97	2.48	3.99	5.16	5.75	5.15	4.09	2.77	1.26	0.45	0.24	(44)	
	RadStd		0.05	0.16	0.28	0.46	0.45	0.27	0.61	0.43	0.38	0.21	0.08	0.02	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.44	N/A	N/A	N/A	N/A	N/A	N/A	-149	N/A	N/A	(46)
(47)	Regional (56 neighbors)	+0.42	N/A	N/A	N/A	N/A	N/A	-131	-158	N/A	N/A	(47)

Nomenclature: See separate page